SIMULATED PHISHING ATTACK USING SHELLPISH, PHP and NGROK

Abstract

Phishing attacks continue to be one of the most widespread cybersecurity threats, targeting users through deceptive techniques. This paper presents a simulated phishing environment using the Shellphish toolkit alongside PHP and Ngrok, with the main goal main of creating awareness around phishing and social engineering attacks and teach the users on how to detect and defend themselves against such attacks. Our project was conducted in a secure virtual lab environment and replicated phishing scenarios to illustrate how attackers collect credentials and location-based information. The results highlight the effectiveness of such tools and emphasise the importance of continuous security training for users.

I. Introduction

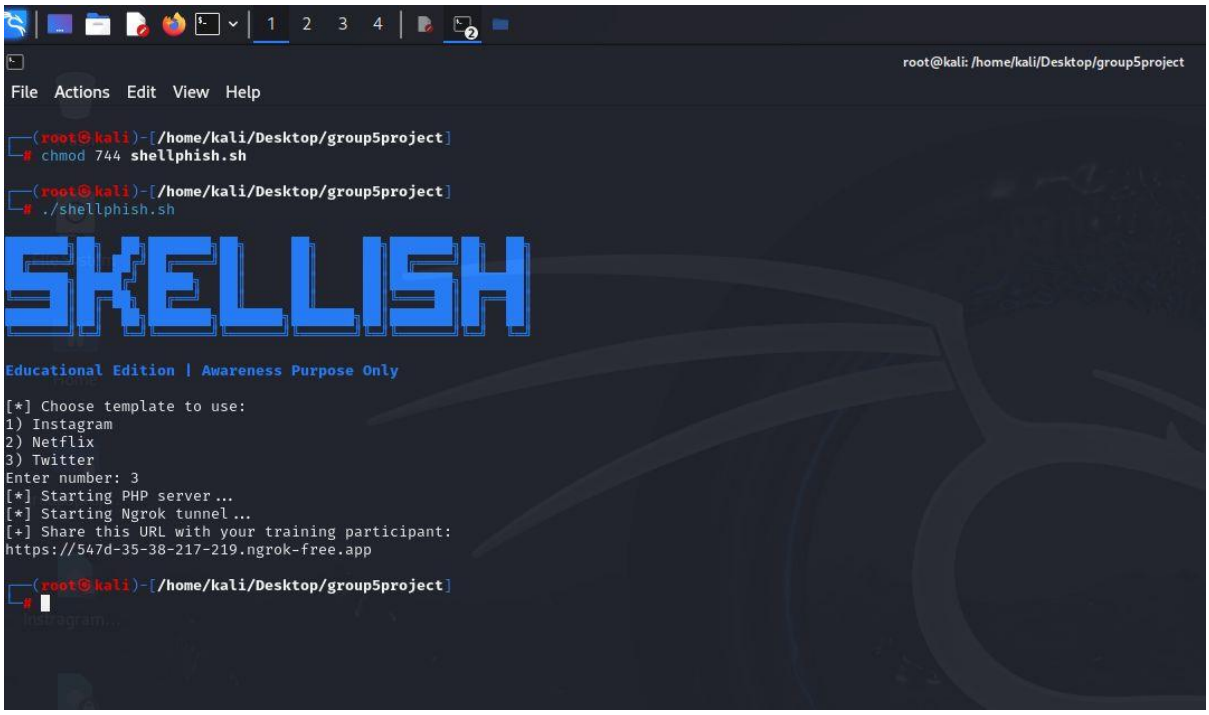
Cyber threats, particularly phishing aims to trick the recipient into falling for the attacker's desired action, such as revealing financial information, system login credentials, or other sensitive information. Fundamentally, these threats exploit human psychology rather than technical vulnerabilities.

As a popular form of social engineering, phishing uses psychological manipulation and deception whereby threat actors masquerade as reputable entities to mislead users into performing specific actions. These actions often involve clicking links to fake websites, downloading and installing malicious files, and divulging private information, like bank account numbers or credit card information.

II. Tools and Technology used

To replicate phishing attacks, we employed the following tools:

- **Shellphish** – A toolkit used to create fake login interfaces for well-known platforms like Twitter, Instagram, Netflix and other webistes.
- **Ngrok** – Enabled secure exposure of local services to the public internet, facilitating real-time phishing simulations.
- **PHP** – Used for handling and storing captured data submitted through the mock login pages.
- **Kali Linux** – Provided a virtualized lab setup for ethical and controlled testing.



III. Methodology

A. Environment Configuration

We began by configuring a virtual machine with Kali Linux and cloning the Shellphish repository. Once set up, Ngrok was integrated to make our phishing pages accessible online without exposing the host system.

B. Simulated Login Pages

Pre-built templates within Shellphish were customized and deployed to mimic the design of popular platforms. These pages were then used to collect test data during the simulation.

C. Email Simulation

Mock phishing emails were created with embedded Ngrok links redirecting to the spoofed login interfaces. Recipients of these emails were encouraged to interact with the links to simulate how real victims might behave.

New Login Detected

Hi there,

We noticed a new login to your Twitter account:

- **Device:** Firefox on Windows
- **Location:** Accra, Ghana
- **IP Address:** 102.64.XX.XX
- **Time:** April 13, 2025 – 9:03 PM GMT

If this was you, you can safely ignore this email.

If not, we recommend securing your account immediately:

Secure My Account

Twitter, Inc. © 2025

Instagram

New Login Attempt

Hello,

We noticed a login attempt to your Instagram account from a new device or location:

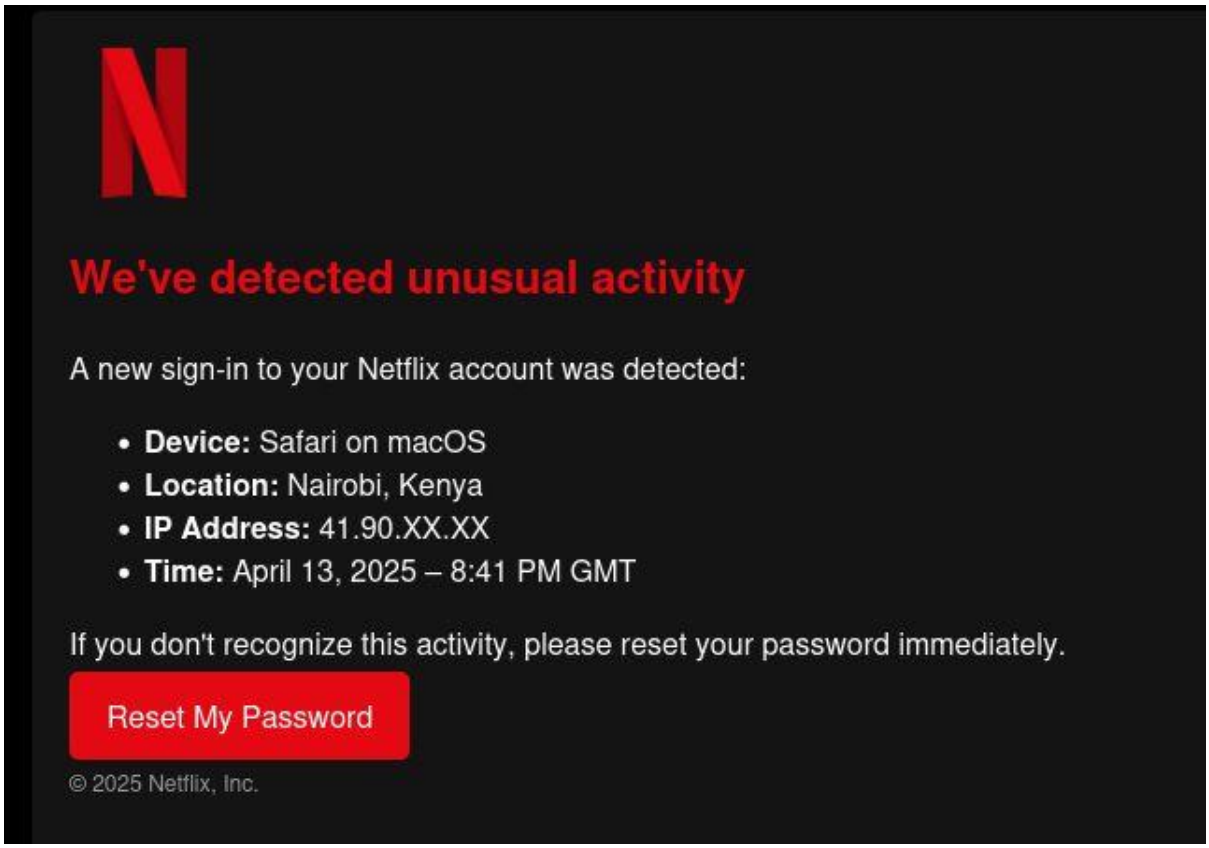
- **Device:** Chrome on Windows
- **Location:** Lagos, Nigeria
- **IP Address:** 197.210.XX.XX
- **Time:** April 13, 2025 – 6:24 PM GMT

If this was you, you can safely ignore this email.

If you don’t recognize this activity, please secure your account immediately:

Secure My Account

This email was sent to you for security purposes. © Instagram Meta Platforms, Inc.

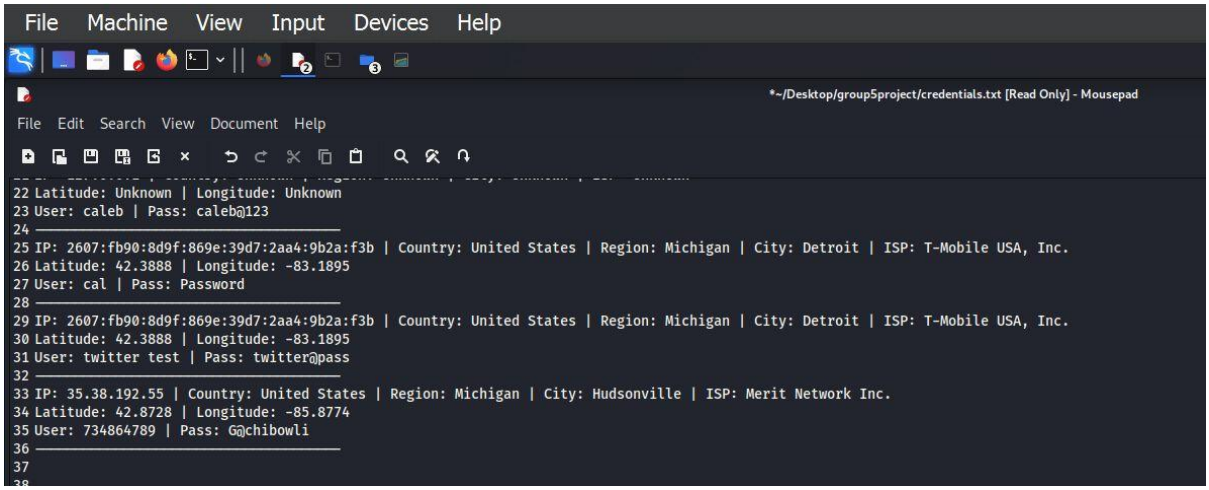


IV. Data Collection and Results

During the simulation, user inputs such as login credentials were captured along with IP-based location information. This data was displayed and stored locally for analysis.

Captured Information Included:

- Username and Password
- IP Address
- Geolocation (City, Region, ISP, Coordinates)



V. Challenges and Mitigations

While working on this simulation, we encountered several issues:

- Ngrok's tunneling occasionally failed due to port configuration conflicts, which were resolved by adjusting settings manually.
- Authentication prompts from GitHub posed cloning difficulties, which were bypassed using direct HTTPS cloning methods.
- Making the phishing pages visually realistic while keeping them functionally sound required design iteration

VI. Awareness and Mitigation Strategies

Security measures organizations can take to mitigate phishing attacks on their employees and systems. Security awareness training and education training through real-world examples and exercises help users identify phishing. Here are some other mitigation strategies.

- Make it difficult for attackers to reach your users
- Help users identify and report suspected phishing emails
- Protect your organisation from the effects of successful phishing emails
- Respond quickly to incidents

VII. Conclusion

The simulation successfully demonstrated how phishing attacks can be orchestrated using open-source tools. The ease with which these attacks can be executed reinforces the urgent need for cybersecurity awareness and training. By understanding how phishing works, users are better equipped to recognize and respond to potential threats. All simulations were conducted ethically with no real data involved.

References

[1] Shellphish Tool: https://www.hackingarticles.in/shellphish-a-phishing-tool/

[2] Ngrok: https://ngrok.com/

[3] Kali Linux: https://www.kali.org/

[4] OWASP: https://owasp.org/www-community/attacks/Phishing